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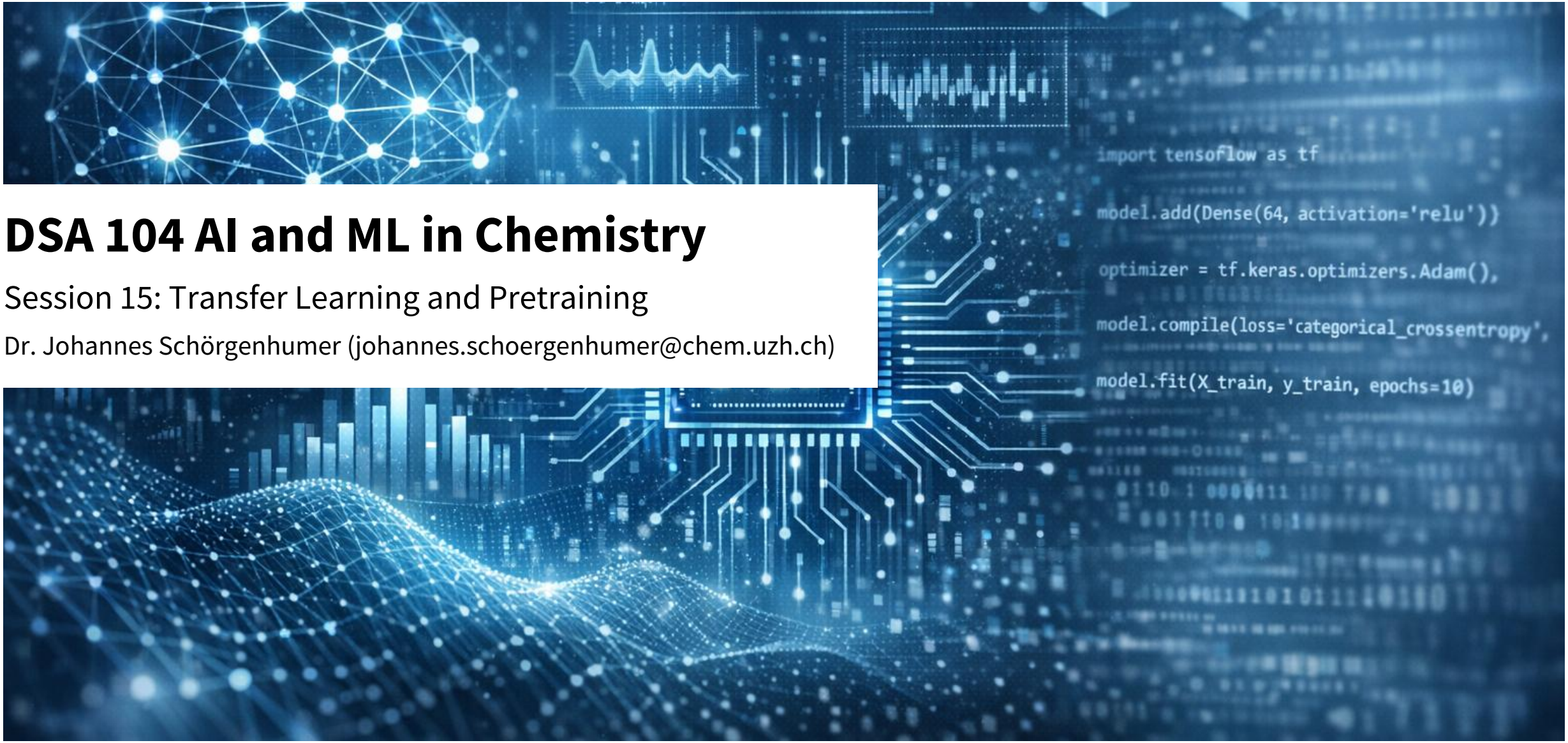

DSA 104 AI and ML in Chemistry

Session 15: Transfer Learning and Pretraining

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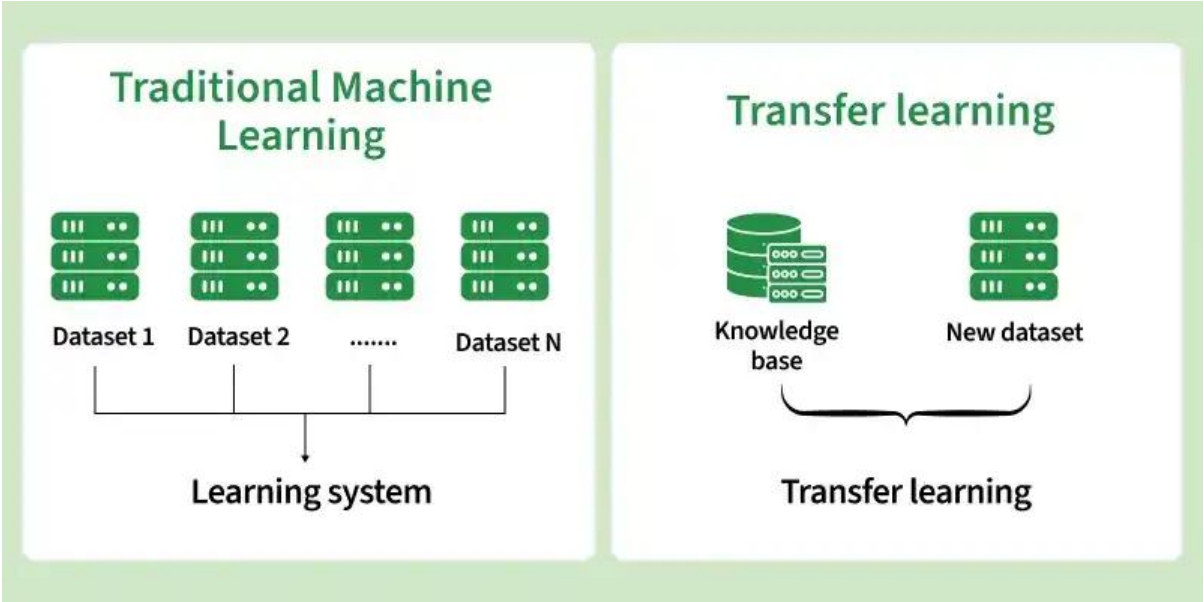
Motivation

- What are problems when training your own models from scratch?
  - Data: Good models are often data-hungry
  - Time: Training and optimising takes a lot of time
  - Expense: Computational time is costly!

- Idea:
  - Reuse knowledge!

- Solution:
  - Use generally pretrained models (e.g. foundation models)
  - Transfer to the specific task / fine-tuning = Transfer Learning



https://www.geeksforgeeks.org/machine-learning/ml-introduction-to-transfer-learning/

Transfer Learning

Definition: Use knowledge from one (general) task/domain and apply to / improve another, usually related task

Components:

— **Source task**: Provides knowledge

— **Target task**: Benefits from that knowledge

— **Transfer mechanism**

Advantages:

— **Fast convergence** during training

— High accuracy with **less data**

— Reduced overfitting on small datasets

— Lower computational cost

Transfer Learning: Types

Inductive Transfer Learning:

— Source and target **tasks are different**

— Target task has labelled data

— Example: ImageNet for medical image classification

Transductive Transfer Learning (Domain Adaptation):

— **Same task, different domains**

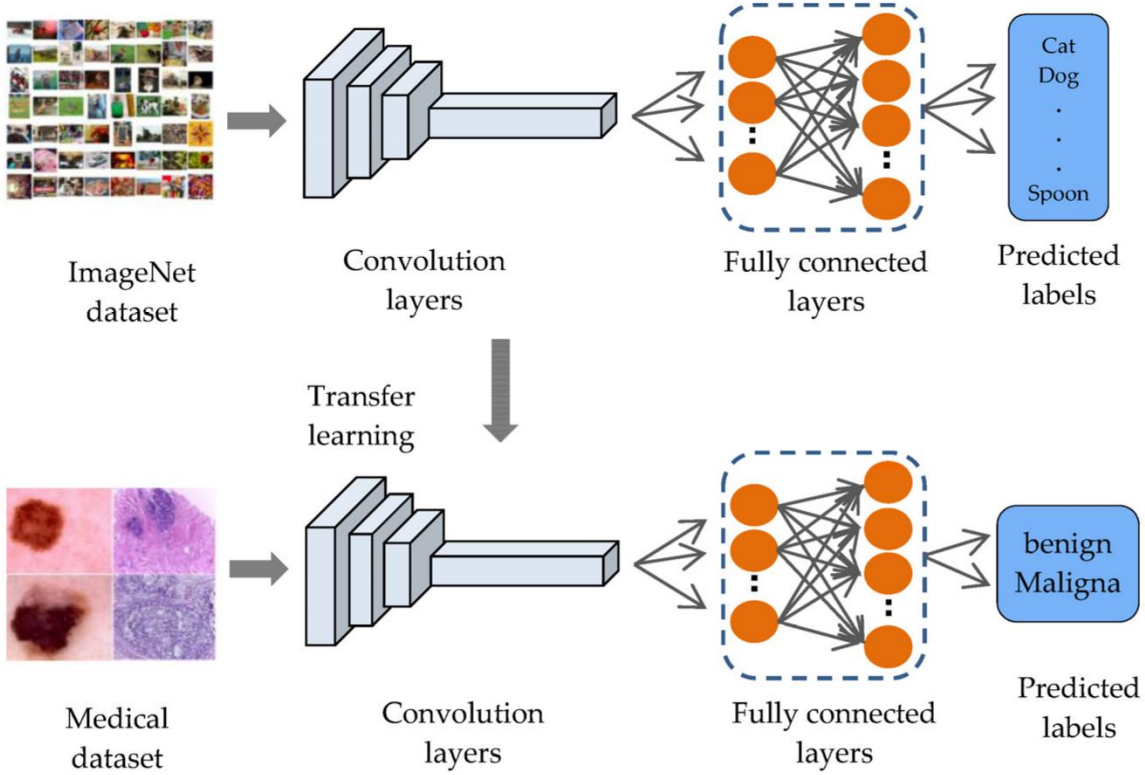
— Little or no labelled target data

— Examples: From product reviews to tweet sentiments, from peer reviewed articles to phenotype classification

Unsupervised Transfer Learning:

— **No labelled data** in either domain

— Example: representation learning across datasets



Sensors **2023**, *23*, 570; https://doi.org/10.3390/s23020570

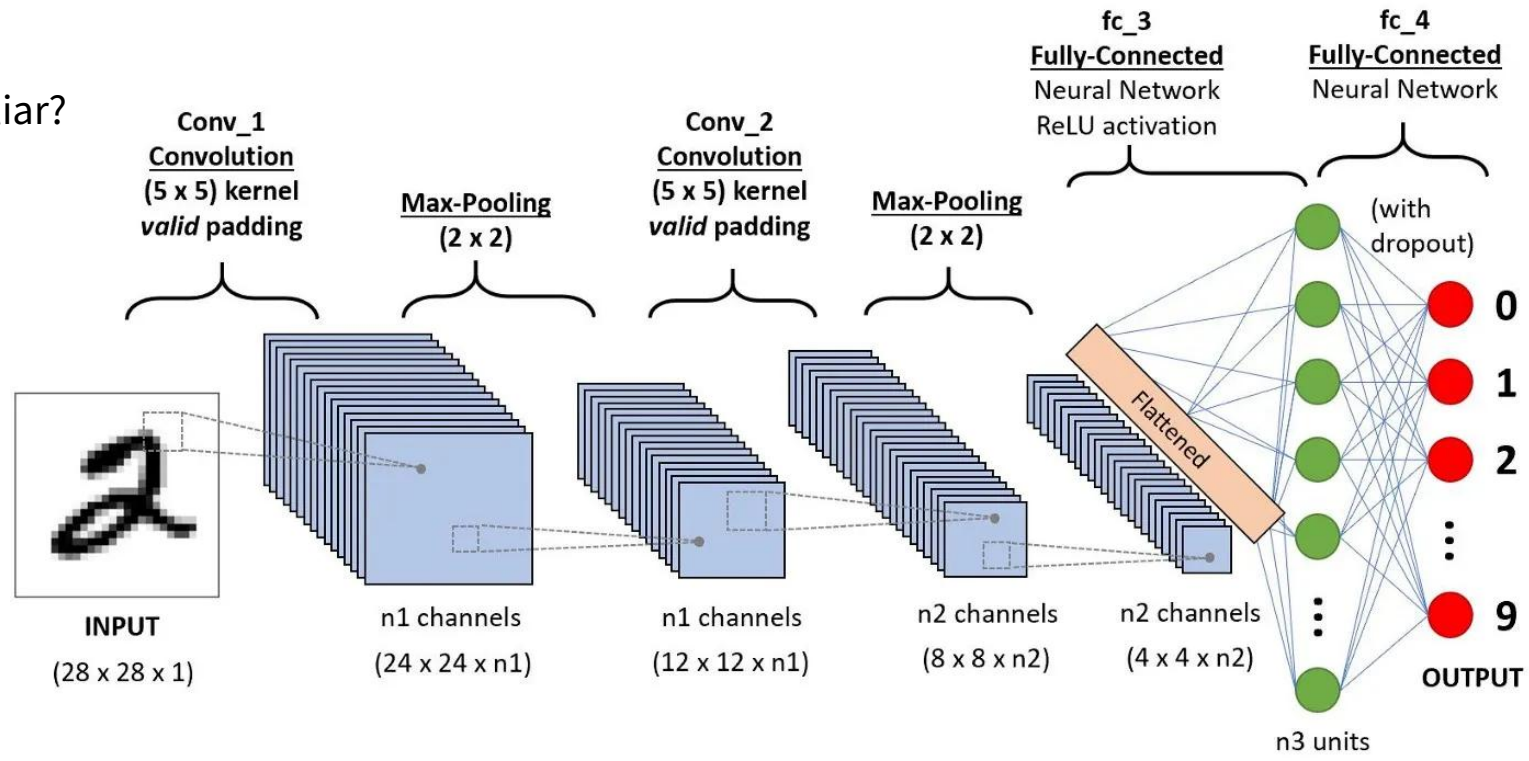
Transfer Learning: How it works

Based on hierarchical representations. Sounds familiar?

E.g. CNNs:

- **Early layers: General features**
  - Images: edges, borders, textures
  - Text: grammar, syntax

- **Later layers: Task-specific features**
  - Images: Objects
  - Text: Sentiment, Intent



Transfer learning **reuses the general representations** / features and **adapts task-specific ones**!

Image from https://medium.com/@mitchhuang777/image-recognition-with-cnns-improving-accuracy-and-efficiency-dd347b636e0c

Common Transfer Learning Strategies: Feature Extraction

Key idea:

— Pretrained model = fixed feature extractor, weights will not be updated

— Only train a new task-specific head (small model)

Pros:

— Can be used on tiny datasets

— Low risk of overfitting

— Cheap and fast

Cons:

— Limited flexibility and domain adaptation

Example: solubility prediction

1) Train a GNN on molecular data: converts molecules into vector embeddings
2) Train a small model (Lin. Reg., small MLP) on top for solubility prediction

Architecture (e.g. solubility prediction)

Molecule (graph)
↓
Pretrained GNN encoder (**frozen**)
↓
Molecular embedding
↓
New regression head (**trainable**)
↓
Solubility prediction

What benefits does this offer over just running some linear model?

Common Transfer Learning Strategies: Fine-tuning

Key idea: Pretrained encoder reused!

— Initialise model with pretrained weights

— Continue training of some or all layers

Variants:

— Fine-tuning only top layers

— Gradual unfreezing

— Full fine-tuning (all layers retrained)

Pros:

— More accuracy

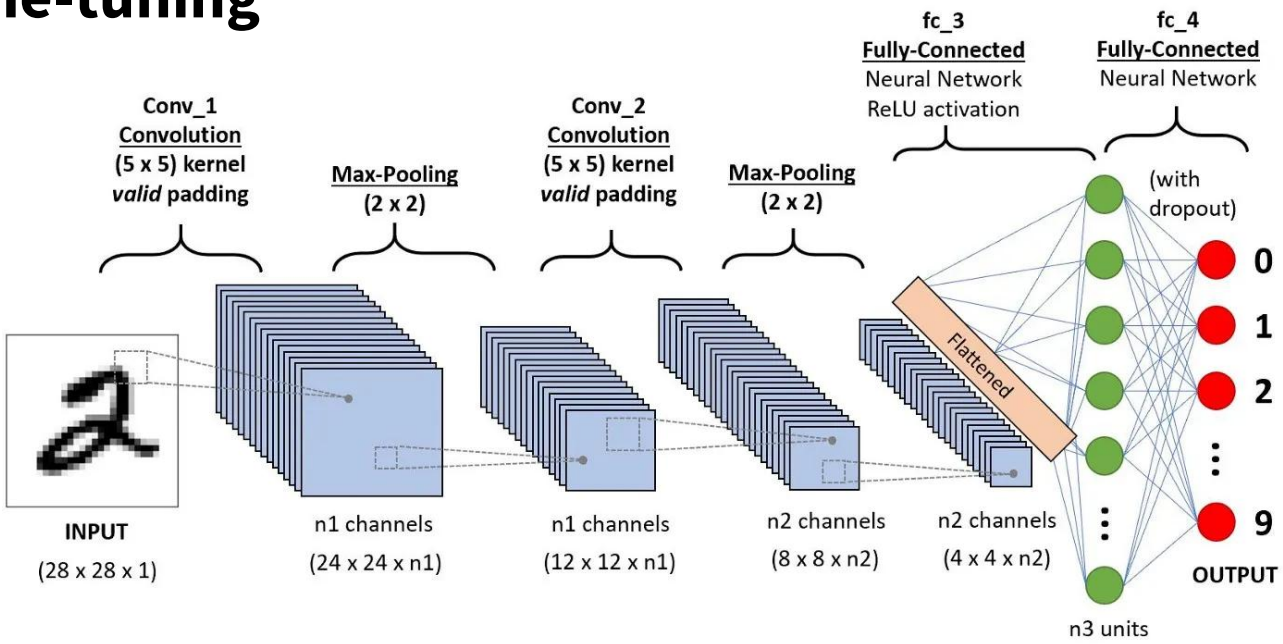
— Higher flexibility

— Strong domain adaptation

Cons:

— More risk of overfitting

— More data needed



Architecture (e.g. binding affinity)

Ligand graph, protein sequence/structure

↓

Pretrained encoder (**trainable**)

↓

Task-specific head (**trainable**)

↓

Binding affinity

Image from https://medium.com/@mitchhuang777/image-recognition-with-cnns-improving-accuracy-and-efficiency-dd347b636e0c

Common Transfer Learning Strategies: Some guidelines

Which approach to choose?

— <1000 samples: feature extraction

— >>1000 of samples: fine-tuning

On fine-tuning: reduce overfitting & allow specialisation, while not drifting away from pretrained model:

— **Partial fine-tuning**: Only unfreeze…

  — Last GNN layers

  — Interaction layers

  — Attention blocks

— Layer-wise **learning rates**:

  — Encoder: very small LR

  — Head: larger LR

— **Regularisation**:

  — Weight decay, Elastic Weight Consolidation (EWC)

  — Early stopping

Other related learning strategy: Joint learning

Core idea:

— **Simultaneous** multi-task learning instead of sequential learning on pretrained models

— All learned **tasks influence each other** during training (combined loss!)

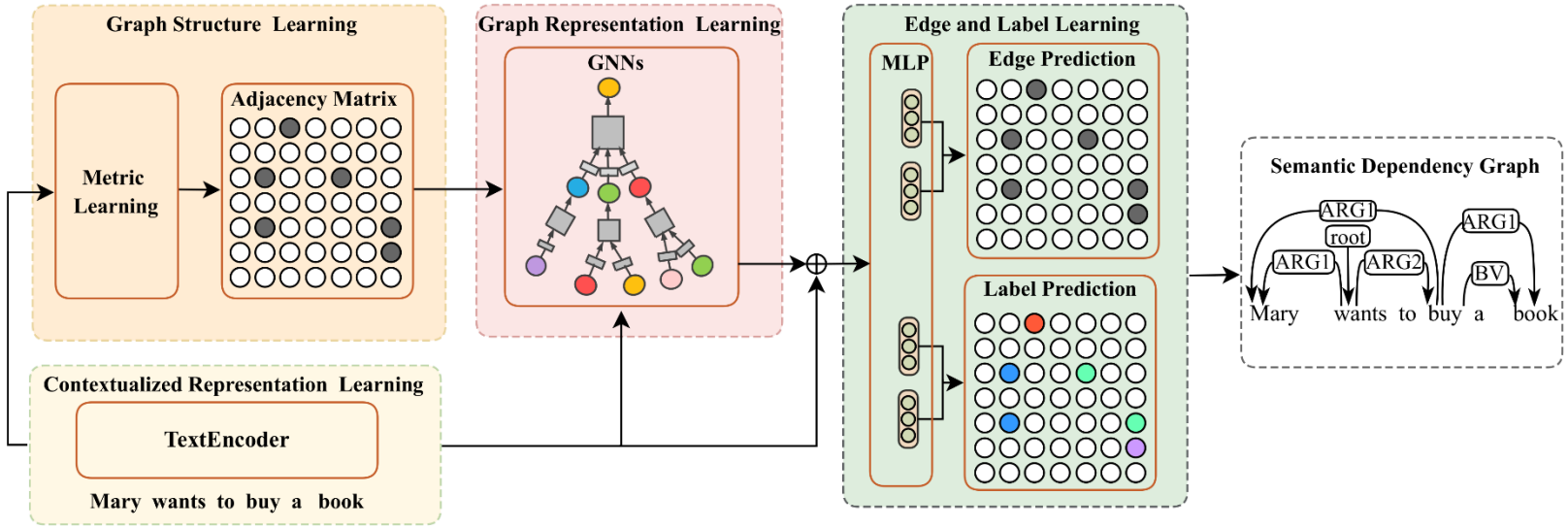
— **Shared structures** across domains can be learned

Somewhere between:

— multi-task learning: **all tasks equally important**

And

— transfer learning: one primary task, auxiliary tasks for training (**implicit regularisation**)



Li, B.; Fan, Y.; Gao, M.; Sataer, Y.; Gao, Z. A Joint-Learning-Based Dynamic Graph Learning Framework for Structured Prediction. *Electronics* **2023**, *12*, 2357. https://doi.org/10.3390/electronics12112357

Other related learning strategy: Joint learning

Application in transfer learning:

— **Shared layers** (encoders, feature extractors) learn cross-domain features

— **Task-specific heads** specialise on each task, but influence each other

— **Gradients from all tasks** update the shared layers jointly

Goal: learn domain-invariant features

Pros:

— Better generalisation

— Implicit regularisation

Challenges:

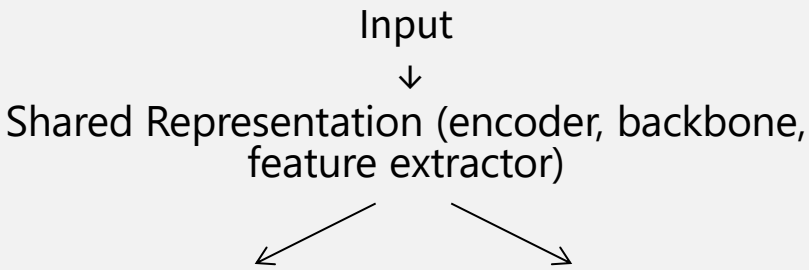
— negative transfer

— task imbalance

— complexity (conflicting gradients)

Example: synthetic and real images

Example architecture:



Transfer Learning Strategies

Imagine the following scenario:

You want to predict specific binding activity between small molecules and a specific human protein target (e.g. a kinase involved in cancer).

You have a lot of binding data for many other proteins (e.g. ChEMBL, > 50 k samples), including ligand-protein interactions, but only a very small dataset (300 samples) on high-quality binding measurements for your protein of interest (your own research results)

For this scenario, **compare classic transfer learning and transfer-oriented joint learning**:

— Create a schematic architecture for each approach.

— Which classic transfer learning type would be applicable?

— Discuss what is learned in the source and the target task

— Discuss the limitations or expected challenges

— Discuss the benefits of each approach

— Which one would you apply?

Challenges in transfer learning

- **Negative transfer:**
  - Transfer learning can hurt performance when:
    - Source and target domains are too dissimilar (**domain shift**)
    - Learned features are misleading for the target task

- **Limited** target domain **data**

- **Mismatch** in task and representation

- **Catastrophic forgetting:**
  - During fine-tuning, model forgets the pretrained knowledge from the source task
  - Causes:
    - Target dataset too small
    - Too high LR
    - Too many layers unfrozen

Transfer Learning: Synopsis

Context of **Deep Learning**:

— **Foundation models** (LLMs, vision models) rely almost entirely on transfer learning

— Fine-tuning, adapters, and LoRA are all transfer mechanisms

— **Zero-shot** and few-shot learning are extreme forms of transfer learning

Takeaways:

— Transfer learning works because representations “matter more” than raw data.

— Use transfer learning when:

  — Data is scarce

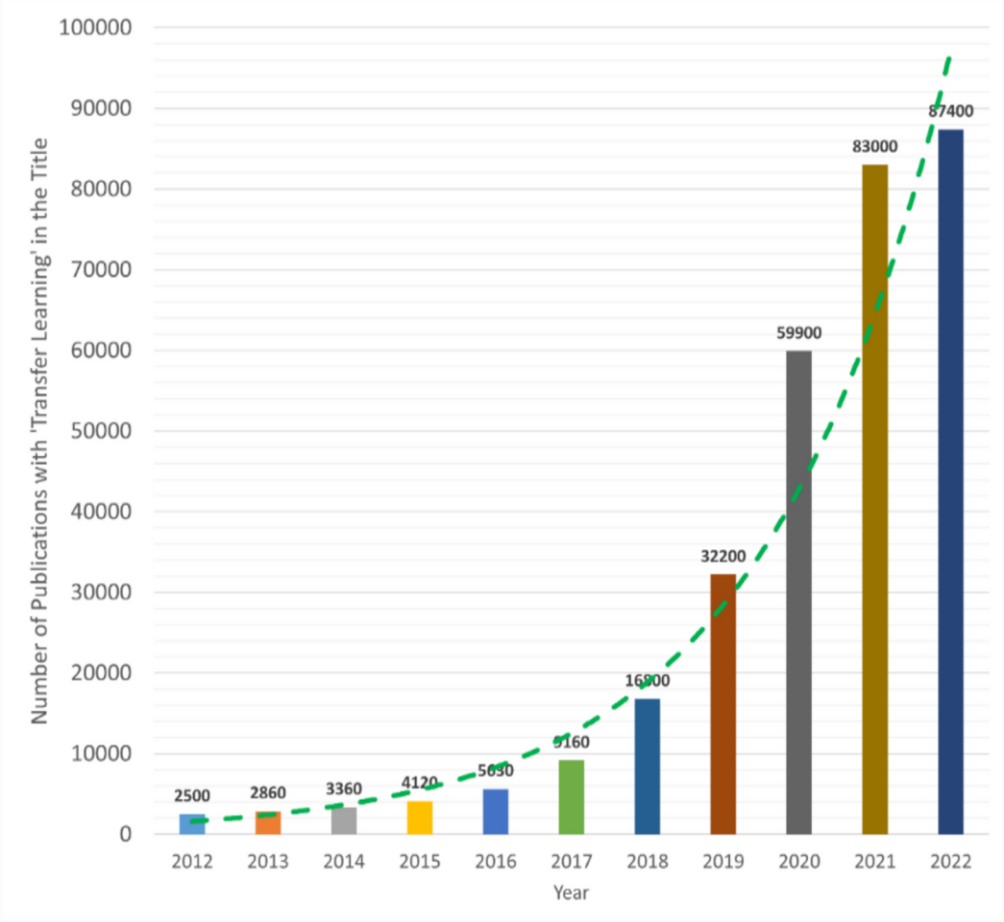
  — Similar pretrained models exist

  — Training from scratch is not feasible

— Be careful when:

  — Domains/tasks are very different

  — Very little target data is available



Gholizade, M., Soltanizadeh, H., Rahmanimanesh, M. *et al.* A review of recent advances and strategies in transfer learning. *Int J Syst Assur Eng Manag* **2025**, *16*, 1123–1162. https://doi.org/10.1007/s13198-024-02684-2

Pretraining

Experimental datasets often limited: **Pretrained models essential**!!

Needed: Large, general dataset.

E.g. for reaction prediction or retrosynthesis: **USPTO dataset**

— Public dataset extracted from patent data (1976 – 2016)

— Millions of chemical reactions in textual format

— Coverage of vast chemical space

— Classic benchmark and pretraining data for AI tasks in Chemistry!

Too good to be true? Limitations:

— Bias toward common reactions

— Limited rare chemistry

— Dataset noise

Machine Learning May Sometimes Simply Capture Literature Popularity Trends: A Case Study of Heterocyclic Suzuki–Miyaura Coupling

Wiktor Beker, Rafał Roszak, Agnieszka Wołos, Nicholas H. Angello, Vandana Rathore, Martin D. Burke,* and Bartosz A. Grzybowski*

Cite This: *J. Am. Chem. Soc.* 2022, 144, 4819–4827 | Read Online

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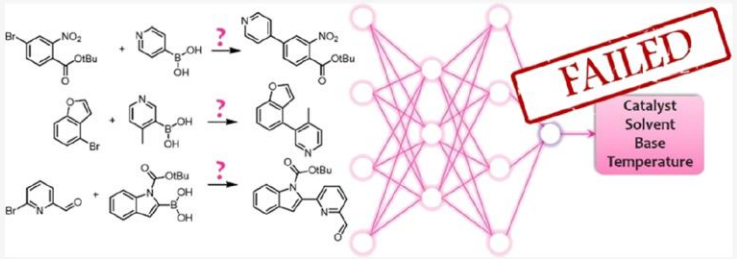


Lowe, D. Chemical Reactions from US Patents (1976-Sep 2016). 2017 https://figshare.com/articles/dataset/Chemical_reactions_from_US_patents_1976-Sep2016_/5104873
Grzybowski et al., *Journal of the American Chemical Society* **2022**, *144*, 4819-4827; DOI: 10.1021/jacs.1c12005

Foundation models

Key idea:

— Large-scale pretrained models

— General-purpose representations

— Adaptable to many tasks

Key Properties:

— Scale (data + parameters)

— Generalisation

— Transferability

Prime example: ChatGPT

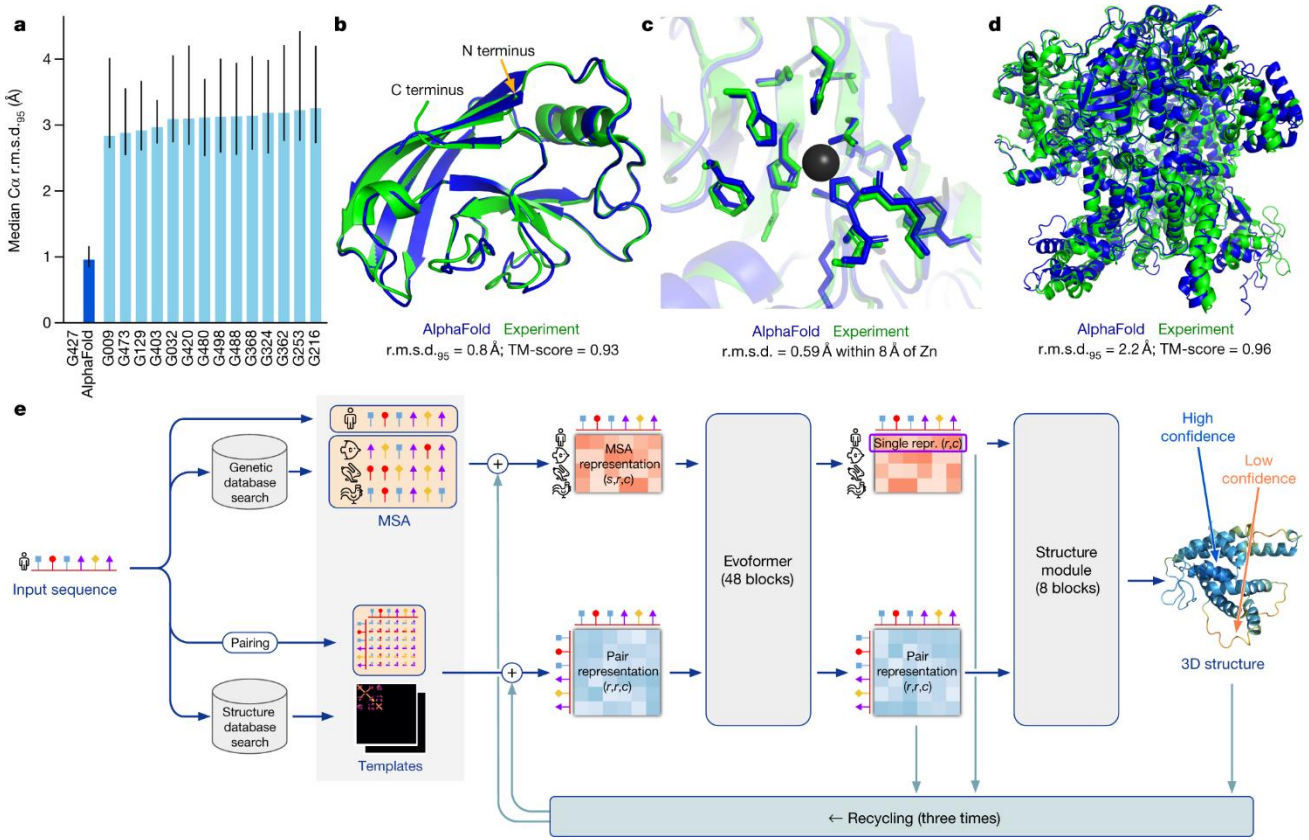
Foundation models - Example: AlphaFold

AI system developed by DeepMind that **predicts the 3D structure of proteins from their amino acid sequences**

— Predicting protein structure: grand challenge in biology

— In CASP14 (2020), AlphaFold achieved near-experimental accuracy for many proteins.

— Accelerated/**disrupted** research in biology, medicine, and drug discovery



Working principle:

— Input = amino acid sequence of a protein

— Uses deep neural networks to learn:

  — Which amino acids are likely to be close in 3D space

  — Geometric constraints (angles, distances)

— Iteratively refines a 3D structure until it converges

— Models spatial relationships directly, not just local patterns

— Treats structure prediction as a geometric reasoning problem

Nature **2021**, *596*, 583-589, https://doi.org/10.1038/s41586-021-03819-2

Foundation models - Example: AlphaFold

What **AlphaFold provides**:

— A **predicted 3D structure**

— A **confidence score**, showing

  — which regions are reliable (likely to be accurate)

  — and which ones shouldn’t be trusted too much (flexible, disordered, uncertain groups)

Used for:

— Understanding protein function

— Drug discovery (target identification, docking)

— Studying diseases caused by misfolded proteins

— Enzyme design, synthetic biology, and basic research

Revolutionary, but still **limited**:

— Dynamic nature of proteins

— Struggles with protein-protein interactions, ligands, metals, post-translational modifications

Foundation models - Example: Uni-Mol

3D-aware foundation model learning rich molecular representations for chemistry and drug discovery tasks, developed 2022 by DP Technology

Uni-Mol learns **general purpose embeddings**:

— Small molecules

— Protein pockets (binding sites)

Working principle:

— Represents **molecules as 3D point clouds** (Atoms + coordinates)

— Uses a transformer architecture adapted for 3D geometry

— Learns from:

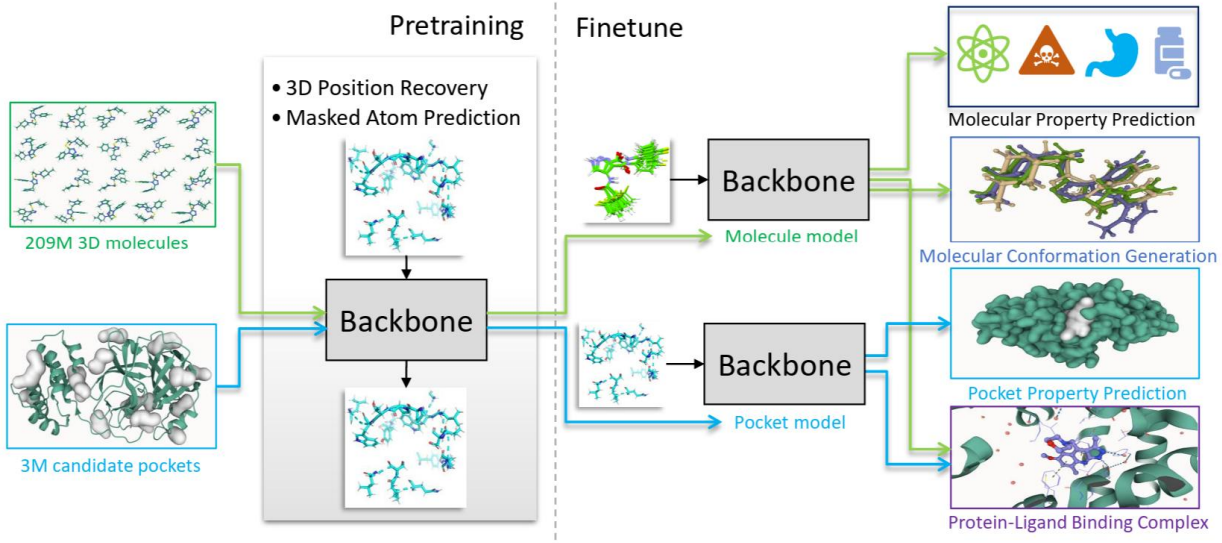
  — **Distances** between atoms

  — **Spatial relationships**

— Is pretrained using self-supervised objectives, such as:

  — Masking atom types or positions

  — Predicting geometric features



https://deepwiki.com/deepmodeling/Uni-Mol
Zhou et al., Conference paper at ICLR **2023**, https://openreview.net/forum?id=6K2RM6wVqKu

Foundation models - Example: Uni-Mol

Output – depending on task:

— Vector embeddings (for ML tasks)

— Predicted properties (e.g. solubility, toxicity)

— Scores for ligand–target compatibility

Typical applications:

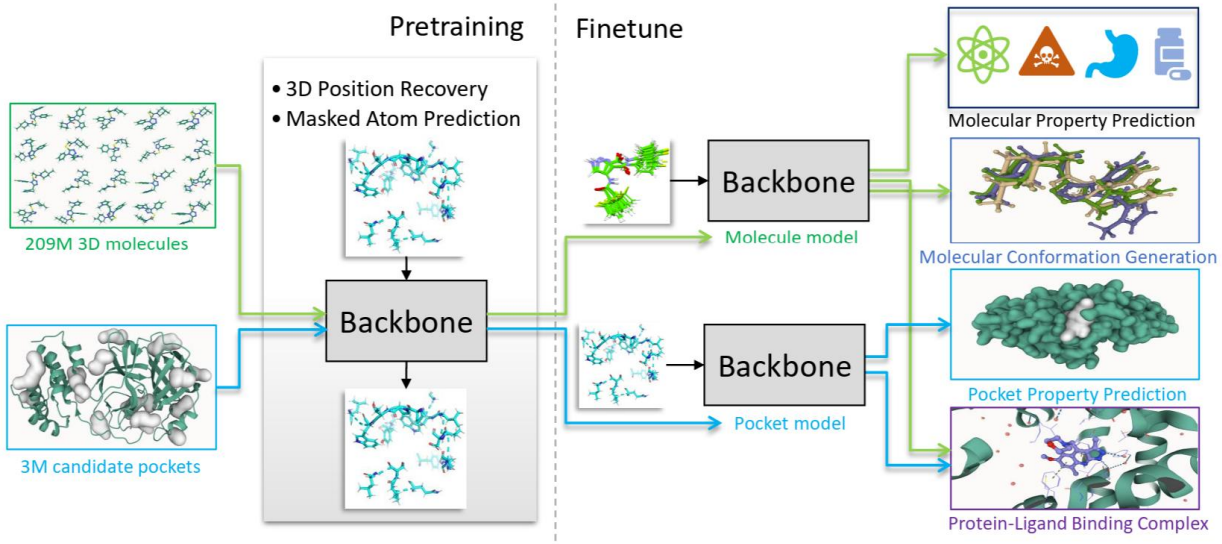
— Drug discovery (Virtual screening, lead optimisation)

— Property prediction

Limitations:

— Requires 3D conformations (or good conformer generation)

— Not designed to predict structures from scratch



https://deepwiki.com/deepmodeling/Uni-Mol
Zhou et al., Conference paper at ICLR **2023**, https://openreview.net/forum?id=6K2RM6wVqKu

Using foundation models

Preparation:

1) In VSC code, after starting the environment, run: *uv remove transformers –optional transformers*
2) Then, run: *uv add transformers --optional tranformers* and *uv sync --extra transformers*
3) Tell me the version number (need to specify the lowest in the project.toml in the repo)

Exercise:

— Use a pretrained foundation model as a fixed encoder to generate features from SMILES in the ESOL dataset.

— Build a small regression model on top to predict solubility

— Compare the performance with small ML models trained on fingerprints or descriptors

Some useful foundation models

Model	Domain	Core idea
AlphaFold, AlphaFold2	Structural biology	Predicting structures based on sequence
ESM (Evolutionary Scale Modelling)	Protein language modelling	Protein sequences as “biological language”, learns embeddings encoding structure, function, and evolutionary constraints; e.g. for structure prediction, mutation effect prediction, protein-design.
Geneformer	Functional genomics	Treats gene expression profiles like sentences, e.g. for Zero-shot cell type annotation, disease gene priorisation
scGPT (single-cell GPT)	Single-cell multi-omics	“GPT for cells”, learns joint cell and gene embeddings. Used for Cell-type annotation, genetic perturbation prediction, …
UCE (Universal cell embedding)	Cell biology	Trained on millions of cells across species to produce general purpose embeddings, focused on transferability rather than task-specifity
MolBERT	Cheminformatics	Early Proto–foundation model for molecules with BERT architecture (Encoder), trained on SMILES, not so much in use anymore, but many models based on it
ChemBERT, ChemBERT2	Cheminformatics	SMILES as chemical language – transformer pretrained on 77M PubChem molecules. Used for predicting toxicity, molecular properties, …
Uni-Mol	3D structure learning	Transformer learning from 3D data to predict properties (DD!)
MoLFormer	Molecular representations	Large transformer trained on SMILES, performance comparable with GNNs
Grover	Molecular graphs	pretraining on millions of molecular graphs for property prediction and virtual screening